Namaste Taste Cloud Migration Plan

A strategic roadmap for migrating Namaste Taste's infrastructure to AWS — improving scalability, security, and operational efficiency.

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Cloud Service Models Overview

IaaS

Core infrastructure — servers, storage, networking — with customer control over OS and applications.

PaaS

Managed development and deployment environment — build apps without managing underlying infrastructure.

SaaS

Fully managed, web-based applications — no installation, maintenance, or updates required.

Namaste Taste will primarily leverage **SaaS**, supported by selective **PaaS** use, to simplify operations and enable scalable, cost-effective solutions.

Service Migration Diagram

Overview of all services being migrated from on-premises infrastructure to AWS cloud environments.

Database Migration: Amazon RDS

Why Migrate?

On-premises SQL databases are costly, hard to scale, and complex to manage as Namaste Taste grows.

Amazon RDS Solution

Fully managed relational database — automates provisioning, backups, and patching. Supports **Multi-AZ deployments** for high availability and scalability under fluctuating workloads.

Backup & Recovery: AWS Backup Services

AWS Backup

Automated, scalable, and highly durable backup solution reducing risk of data loss.

RDS Automated Backups

Continuously protects relational data with scheduled snapshots and change tracking.

DynamoDB PITR

Point-in-Time Recovery restores databases to an exact moment before corruption or accidental deletion.

Cloud-based backups also support compliance and auditing by enabling access to historical data states.

Directory Services: AWS Managed Microsoft AD

Namaste Taste will implement a **hybrid directory services model** using AWS Managed Microsoft Active Directory — extending on-premises AD into the cloud while preserving existing domain structure, user accounts, and security policies.

High Availability

Domain controllers deployed across multiple Availability Zones with built-in redundancy.

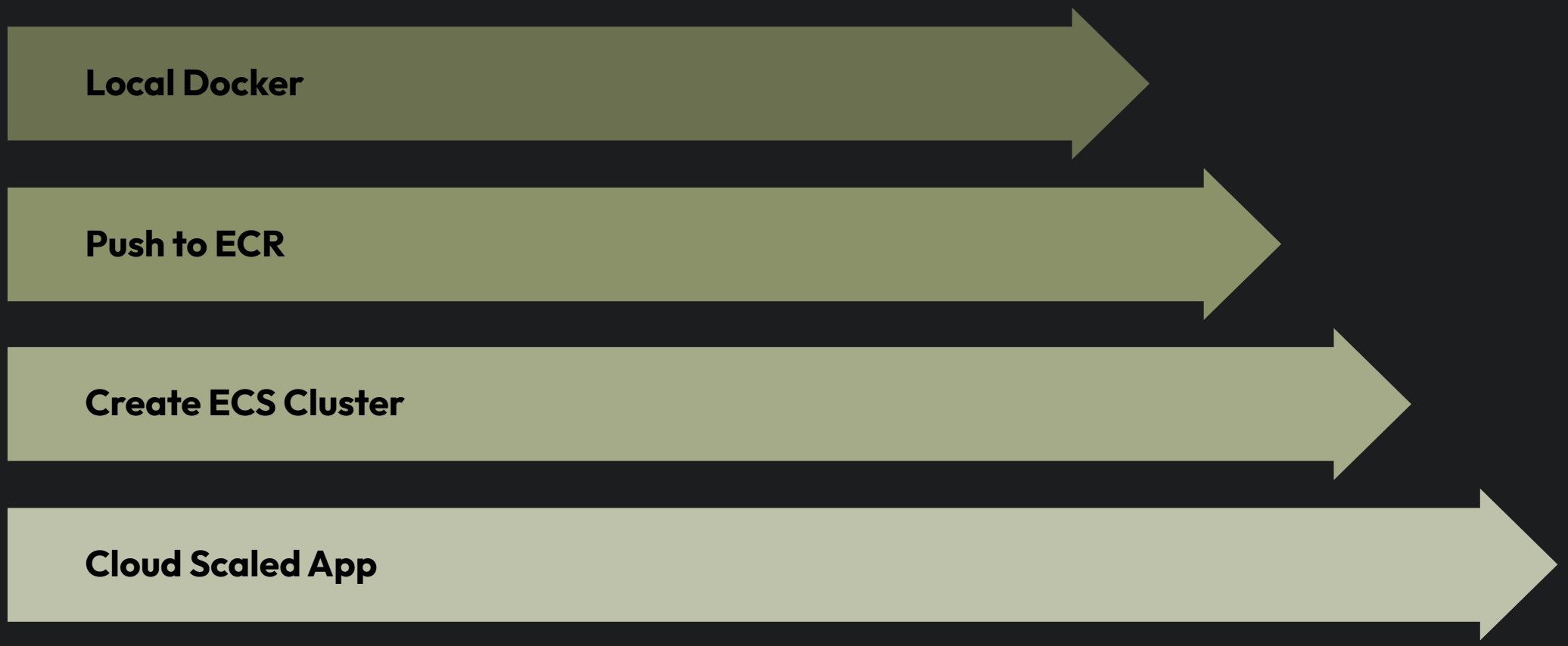
Seamless Sync

On-premises and cloud systems stay synchronized — no disruption to POS, vendor portals, or apps.

Reduced Overhead

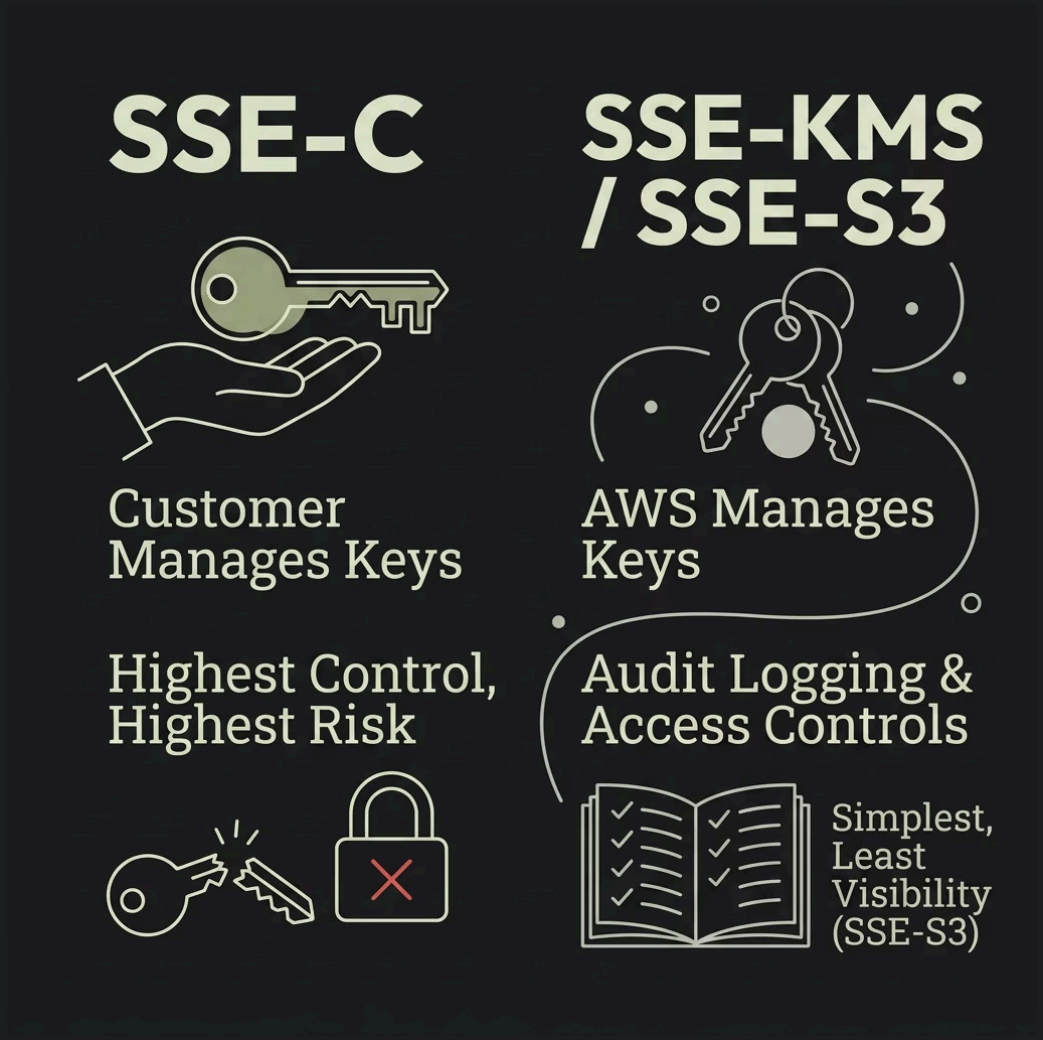
Eliminates physical server management, patching, and manual backup tasks.

Web Application Hosting: Docker & AWS



WordPress was containerized locally using Docker, then deployed to AWS via ECR and container orchestration — enabling high availability, easy updates, and scalability for increased traffic.

Data Security: SSE-KMS Encryption



Recommended: SSE-KMS

AWS Key Management Service provides the best balance of security and control — enabling key creation, access control enforcement, and full audit logging for compliance.

- SSE-KMS aligns with industry standards for protecting customer, payment, and employee data.

Network Security: VPC, Security Groups & NACLs

Security Groups

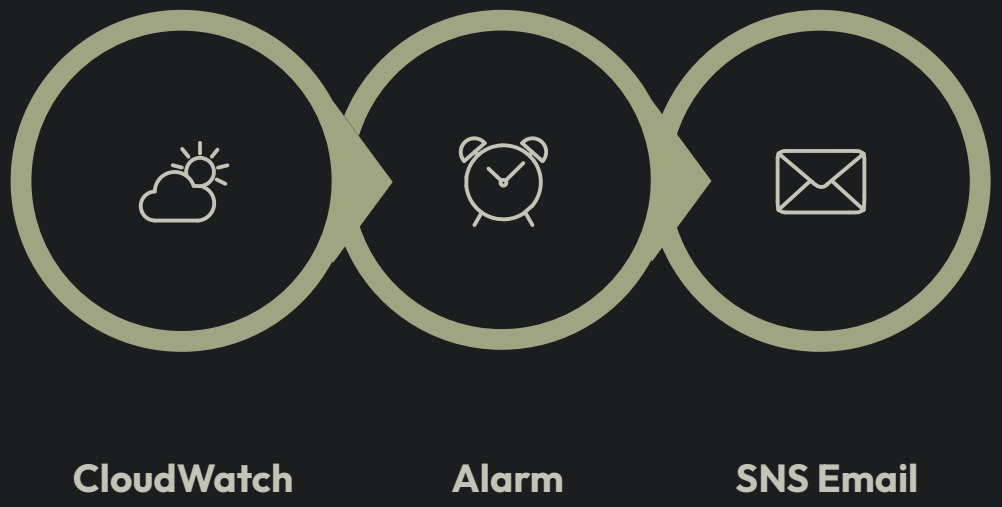
Stateful — operate at the resource level. Inbound traffic denied by default; return traffic automatically allowed.

NACLs

Stateless — operate at the subnet level. Both inbound and outbound rules must be explicitly defined and evaluated in order.

This layered approach filters traffic at multiple levels, reducing unauthorized access risk across Namaste Taste's cloud environment.

Cloud Monitoring: CloudWatch & SNS



Automated Alerting

CloudWatch alarms trigger when CPU usage exceeds defined thresholds (e.g., 50%), sending SNS notifications to administrators via email.

Tested with a CPU stress tool — alarm detected elevated usage and returned to normal once performance stabilized.

Hybrid Connectivity: On-Premises to AWS

During migration, on-premises systems remain operational while integrating with AWS resources — ensuring business continuity. Administrators connect remotely via **SSH over public IPv4** to manage and configure cloud instances.

Flexible Scaling

Instance types can be resized to meet changing workload demands without disrupting operations.

Gradual Transition

Workloads migrate incrementally, reducing risk and ensuring seamless integration between environments.

Cloud-Based POS System Architecture

API Gateway

Handles communication between POS devices (tablets/registers) and AWS backend services.

AWS Lambda

Processes transactions, orders, and updates serverlessly — no dedicated servers required.

RDS / DynamoDB

Stores orders, payments, and employee data with high availability and reliability.

S3 + KMS

Stores receipts and reports; KMS encrypts sensitive customer and payment data.

POS Monitoring & Reliability

CloudWatch & SNS Integration

Amazon CloudWatch and SNS provide real-time performance tracking and alerts for the POS system — ensuring high transaction volumes are handled efficiently and issues are detected proactively.

This architecture eliminates on-premises POS infrastructure while supporting future growth and system integrations.

Backout Plan: Triggers

The backout plan activates if any of the following occur during migration:

Critical Service Failures

Database corruption, failed POS transactions, website downtime, or failed user authentication.

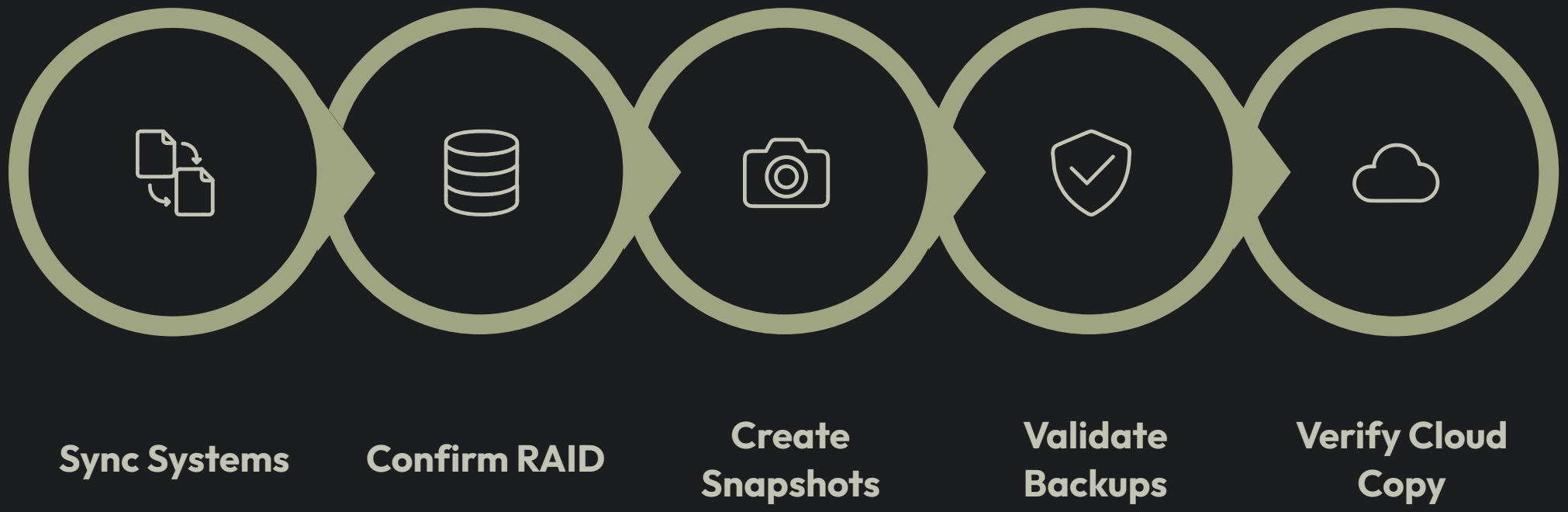
Excessive Downtime

Any service unavailable beyond the approved downtime window.

Failed Testing

Migrated systems not functioning as expected during validation.

Backout Plan: Pre-Migration Safeguards



These steps ensure a reliable, known-good version of the environment exists before any major changes are made.

Backout Plan: Recovery Steps

01

Data Corruption

Replace corrupted data with verified backup data immediately.

02

Application Failure

Reinstall application on local server and restore using backup data.

03

Storage Failure

Initiate RAID restoration to rebuild redundancy.

04

Sync Issues

Disconnect cloud sync and initiate reverse data flow to restore on-premises environment.

Migration Summary: AWS Services at a Glance

Service	AWS Solution	Key Benefit
Database	Amazon RDS	Managed, scalable, Multi-AZ
Backups	AWS Backup / PITR	Automated, point-in-time recovery
Directory	AWS Managed Microsoft AD	Hybrid identity management
Web Hosting	Docker + ECR + ECS	Containerized, scalable
Security	SSE-KMS	Encrypted, auditable
Network	VPC + Security Groups + NACLs	Layered protection
Monitoring	CloudWatch + SNS	Proactive alerting
POS	API Gateway + Lambda + RDS	Serverless, real-time

Key Benefits of Cloud Migration

Scalability

Resources scale on demand to match business growth and traffic fluctuations.

Security

Layered encryption, access controls, and audit logging protect sensitive data.

Cost Efficiency

Eliminates physical hardware costs and reduces administrative overhead.

Reliability

Multi-AZ deployments and automated backups ensure high availability and continuity.

Looking Ahead

By adopting a **SaaS-first, PaaS-supported** cloud strategy on AWS, Namaste Taste is positioned for sustainable growth — with a secure, scalable, and modern infrastructure that supports both current operations and future expansion.

> The hybrid connectivity model ensures a smooth, low-risk transition — keeping the business running while the future is built in the cloud.